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Predicting Fuel Efficiency Using Regression Trees

Graded Quiz • 15 min

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2.1 Data Representation

2.2 Statistical Data Analysis

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2.4 Dimensionality Reduction

2.5 Data Classification

2.6 Data Prediction

Video: Data Prediction

56 min

Graded App Item: EstimatorErrorPlot

15 min

Quiz: Predicting Fuel Efficiency Using Regression Trees

Submitted

Quiz: Predicting Fuel Efficiency Using Gaussian Process Regression

Started

● Congratulations! You passed!

Grade received 100%

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Predicting Fuel Efficiency Using Regression Trees

1. Load the 'carbig' dataset into MATLAB. Curate the data into training and testing set using the following code:1 / 1 point

```
1 load carbig
2 D = [Acceleration, Cylinders, Displacement, Horsepower, Model_Year, Weight];
3 y = mpg;
4 msk = ~isnan(y);
5 D = D(msk,:);
6 y = y(msk);
7
8
9 D_train = D(1:2:end,:);
10 y_train = y(1:2:end);
11
12 D_test = D(2:2:end,:);
13 y_test = y(2:2:end);
14 rng(8);
```

Submit your assignment

Due Apr 21, 11:59 PM IST

Receive grade

To Pass 50% or higher

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Your grade 100%

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Fit a regression tree to the training dataset (D_train, y_train) using the automatic parameter optimization option.

What is the name of the function we use to train a regression tree?

fitrtree

Correct

2. Use the model with the predict function to estimate fuel efficiency for cars in the testing dataset (D_test).1 / 1 point

Compare the estimation with the true values (y_test). What's the mean squared difference between the two, accurate to two decimal places? You can use the EstimatorErrorPlot function we made to find this.

9.15

Correct